

EE708 - Assignment 3

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Q1 (a): If s is a metric similarity measure on a set X , prove that $s(x, y) + a$ is also a metric similarity measure for $a \geq 0$.

Let $s(x, y) \geq 0$ for all $x, y \in X$.

Define:

$$s'(x, y) = s(x, y) + a$$

where $a \geq 0$.

1. Non-negativity:

$$s'(x, y) = s(x, y) + a \geq 0$$

Since both terms are non-negative.

2. Symmetry:

If

$$s(x, y) = s(y, x)$$

then

$$s'(x, y) = s'(y, x)$$

So properties remain same.

Hence, $s'(x, y)$ is also a metric similarity measure.

(b): If d is a metric dissimilarity measure, prove $d(x, y) + a$ is also metric.

Answer:

Let

$$d'(x, y) = d(x, y) + a$$

1. Non-negativity:

$$d'(x, y) \geq 0$$

2. Symmetry:

$$d'(x, y) = d'(y, x)$$

3. Triangle Inequality:

Given:

$$d(x, z) \leq d(x, y) + d(y, z)$$

Adding a does not change inequality.

Therefore, $d'(x, y)$ is also metric.

Q2: Prove Euclidean distance satisfies triangle inequality.

We want to prove:

$$\|x - z\| \leq \|x - y\| + \|y - z\|$$

Let:

$$a = x - y$$

$$b = y - z$$

Then:

$$x - z = a + b$$

Now compute square of norm:

$$\|x - z\|^2 = \|a + b\|^2$$

Using dot product:

$$\|a + b\|^2 = (a + b) \cdot (a + b)$$

Expanding:

$$= a \cdot a + a \cdot b + b \cdot a + b \cdot b$$

Since $a \cdot b = b \cdot a$:

$$= \|a\|^2 + 2(a \cdot b) + \|b\|^2$$

Now using Cauchy–Schwarz inequality:

$$|a \cdot b| \leq \|a\| \|b\|$$

So,

$$2(a \cdot b) \leq 2\|a\| \|b\|$$

Therefore:

$$\|a + b\|^2 \leq \|a\|^2 + 2\|a\|\|b\| + \|b\|^2$$

$$= (\|a\| + \|b\|)^2$$

Taking square root on both sides:

$$\|a + b\| \leq \|a\| + \|b\|$$

Substituting back:

$$\|x - z\| \leq \|x - y\| + \|y - z\|$$

Hence proved.

Q3:

Required to check whether $d(x, y) = |x - y|^2$ is a valid metric.

1. Non-negativity:

$$|x - y|^2 \geq 0$$

2. Symmetry:

$$|x - y|^2 = |y - x|^2$$

3. Triangle inequality test:

Let $x = 0$, $y = 1$, $z = 2$.

$$d(x, z) = 4$$

$$d(x, y) + d(y, z) = 1 + 1 = 2$$

Since $4 \not\leq 2$, triangle inequality fails.

Hence, it is **not a valid metric**.

Q4:

Given $x = [6, 4]^T$ and Euclidean distance:

$$\delta(x, v) = \sqrt{(x_1 - v_1)^2 + (x_2 - v_2)^2}$$

Distances:

$$d_1 = \sqrt{(6 - 1.5)^2 + (4 - 1.5)^2} = 5.148$$

$$d_2 = \sqrt{(6 - 2)^2 + (4 - 1)^2} = 5$$

$$d_3 = \sqrt{(6 - 2.5)^2 + (4 - 1.75)^2} = 4.160$$

$$d_4 = \sqrt{(6 - 1.5)^2 + (4 - 2)^2} = 4.925$$

$$d_5 = \sqrt{(6 - 3)^2 + (4 - 2)^2} = 3.606$$

$$d_6 = \sqrt{(6 - 1)^2 + (4 - 3.5)^2} = 5.025$$

$$d_7 = \sqrt{(6 - 2)^2 + (4 - 3)^2} = 4.123$$

$$d_8 = \sqrt{(6 - 3.5)^2 + (4 - 3)^2} = 2.693$$

Single Linkage:

$$D_{min} = 2.693$$

Complete Linkage:

$$D_{max} = 5.148$$

Average Linkage:

$$D_{avg} = \frac{5.148 + 5 + 4.160 + 4.925 + 3.606 + 5.025 + 4.123 + 2.693}{8}$$

$$D_{avg} = 4.335$$

Q5:

The given graph contains two clusters: an elongated cluster $(x_1, x_2, x_3, x_4, x_5, x_6, x_7)$ and a compact cluster $(x_8, x_9, x_{10}, x_{11})$. Using hierarchical clustering, we construct dendrograms using single linkage and complete linkage.

Single Linkage:

Single linkage uses the minimum distance between clusters:

$$D_{\min}(A, B) = \min_{x \in A, y \in B} d(x, y)$$

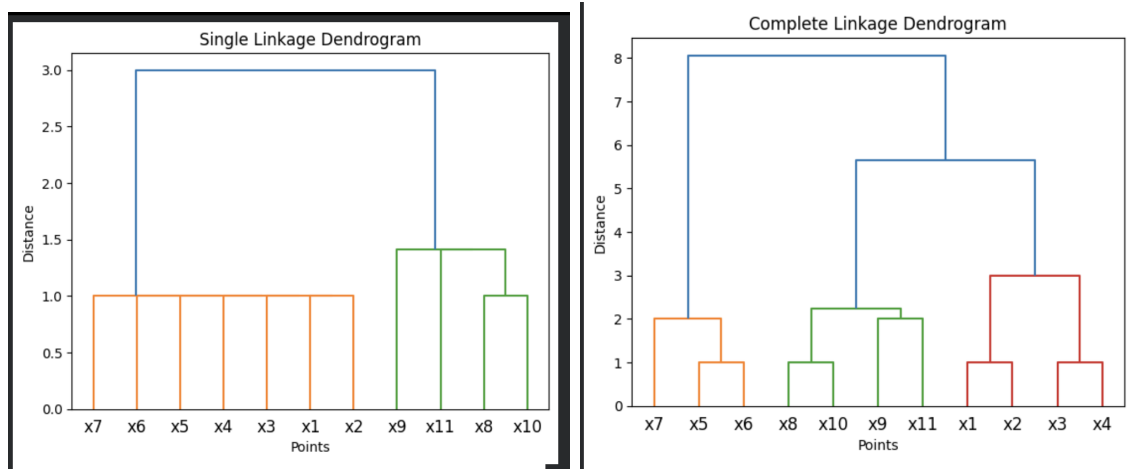
This method often produces chain-like clusters because points can be added sequentially based on the smallest distance.

Complete Linkage:

Complete linkage uses the maximum distance between clusters:

$$D_{\max}(A, B) = \max_{x \in A, y \in B} d(x, y)$$

This produces more compact clusters since it considers the farthest points between clusters.



Observation:

Single linkage creates a chaining effect in the elongated cluster, while complete linkage forms more compact clusters and merges them at larger heights in the dendrogram.

Q6:

In hierarchical clustering, clusters are formed by successively merging the closest clusters based on a linkage criterion. In this question, we apply two linkage methods: single linkage and complete linkage.

Single Linkage

In single linkage clustering, the distance between two clusters is defined as the minimum distance between any pair of points from the two clusters.

$$D_{\min}(A, B) = \min_{x \in A, y \in B} d(x, y)$$

This method tends to form chain-like clusters because clusters can be merged through nearby points.

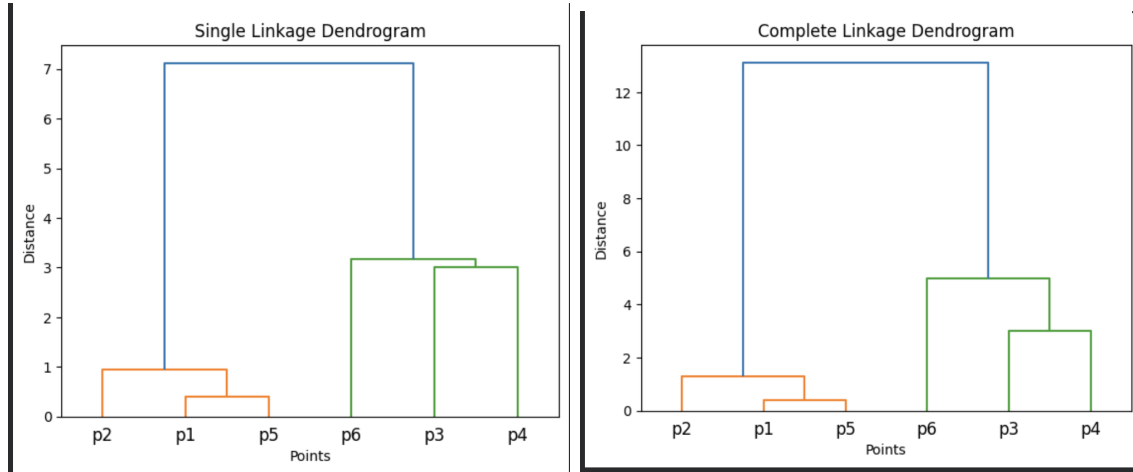
Complete Linkage

In complete linkage clustering, the distance between two clusters is defined as the maximum distance between any pair of points belonging to the two clusters.

$$D_{\max}(A, B) = \max_{x \in A, y \in B} d(x, y)$$

This method generally produces more compact and well-separated clusters.

The dendrograms obtained from applying single linkage and complete linkage algorithms are shown below.



Observation

From the dendrograms, we observe that single linkage tends to merge clusters based on the smallest pairwise distance, which can create elongated or chain-like clusters. In contrast, complete linkage considers the largest distance between clusters, resulting in more compact and balanced cluster structures.

Q7: Can we prune dendrogram?

Yes.

We choose a threshold value and cut the dendrogram horizontally.

Clusters formed below that cut become final clusters.

Q8: How to make K-means robust to outliers?

- K-means is sensitive to outliers because it minimizes squared Euclidean distance mean.
To make it robust:

- Use K-Medoids (PAM): Use an actual data point (medoid) as the center instead of the mean. Medoids are far less affected by extreme values.
- K-Medians: Replace the mean with the median in each dimension and use L_1 (Manhattan) distance.
- Trimming: Implement Trimmed K-means, where a small percentage of points with the largest residuals (distance from their center) are excluded from the centroid calculation in each iteration.
- Preprocessing: Use density-based methods (like DBSCAN) or statistical tests (Z-score) to identify and remove outliers before running the algorithm